

The Dutch East India Company's Eighteenth-Century Workforce: an Enriched Data Collection

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We present a dataset on 460,452 individuals employed by the Dutch East India Company (VOC) in the seventeenth and eighteenth centuries, developed from 774,200 muster records in the 'VOC-opvarenden' collection. The original data has been enhanced through the disambiguation of individual records, the standardization of 44,152 unique place names, and the addition of wage details and rank structure. This paper provides an in-depth overview of the original dataset, the enhancement process, and potential applications. Additionally, it features appendices serving as codebooks, offering concise descriptions of each variable present in the enriched data files. Enabling research into career patterns, network structures, and migration trends, this resource is of significant value to the study of early modern history, social and economic history, and sociology.

Additional Key Words and Phrases: digital humanities, digital history, data processing, social and economic history

1 Introduction

The Dutch East India Company (*Verenigde Oost-Indische Compagnie, VOC*), operational from 1602 to 1800, is remarkable for its extensive activities and modern organizational structure, and notorious brutal conduct. It is also well-known for its exhaustive record-keeping. The Company meticulously documented its widespread activities, sending a vast array of reports, instructions, and ledgers between its Asian trading posts and Batavia (present-day Jakarta), with copies then dispatched to the Netherlands. Of these records, an estimated 25 million pages have survived, mainly in the Netherlands, Indonesia, India, Sri Lanka, and South Africa. This extensive archival treasure is recognized in the UNESCO Memory of the World Register, underlining its historical importance.

A key component of the VOC archives is the ships' pay ledgers, detailing the salaries of seamen and soldiers enlisted for voyages to Asia. The ledgers of 3,248 outbound voyages have survived, mainly dating from the eighteenth century.¹ These ledgers contain 774,200 muster records. Between 2000 and 2012, these records were digitized into the 'VOC-opvarenden' database.^[14] It is hosted by the Dutch National Archives.² Shortly after completion of the dataset, de Boer et al. [4] made attempts to enhance its usability by standardizing place names and integrating related datasets. While their endeavor was commendable, it had limitations due to their exclusive reliance on automated processes, resulting in some quality shortcomings. Our work has focused on editing and enhancing this database with a combination of automated and manual processes, improving its utility

¹The total number of outward voyages undertaken by the VOC and its predecessors is 4,787 (period: 1595-1794)[2]. See also the Dutch-Asiatic Shipping (DAS) database: <https://resources.huygens.knaw.nl/das>

²See <https://www.nationaalarchief.nl/onderzoeken/index/nt00444>. Apart from the Dutch National Archives, the archival institutions of Delft, Rotterdam, Zeeland and West-Friesland were involved in the project, as well as the history departments of the universities of Leyden and Ghent. Employees of Combiwerk Delft did most of the data entry work.

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for research purposes. This paper aims to describe the original data in detail and elucidate the methods used in enriching them, providing a comprehensive guide for researchers utilizing the published dataset.

Since the initial online publication of the muster records, a diverse group of historians, social scientists, and genealogists have extensively utilized this data in their research[1, 6–13, 15–18]. However, its full potential remains largely untapped. To date, researchers have primarily used the dataset for retrieving information on specific individuals or small groups. This limited usage is due to two main challenges in employing the data for more comprehensive and structured analyses of the VOC’s seafaring workforce.

The first challenge with the database is its focus on employments rather than individuals. The original VOC pay ledgers, designed solely for salary calculations, closed employment records at the end of a sailor or soldier’s service. Subsequent re-employments were recorded separately, resulting in fragmented data for those who served multiple times. Consequently, research targeting individual histories necessitates a complex process of clustering related observations to form complete personal profiles.

The second major challenge is the dataset’s limitations for geographical analysis, a key interest area for researchers given the VOC’s highly international workforce. Clerks at the VOC recorded the origins of crew members based on what they thought they heard an employee say and all had their own style of abbreviation, resulting in a wide range of spellings for the same locations. In the digitization project, no standardization of place names was undertaken. The town of The Hague, for example, is represented in 333 different forms in the original dataset. In all, it contains over 150 thousand unique toponyms. Standardizing these place names, essential for any comprehensive geographical analysis of the VOC personnel, is a complex and time-consuming task.

Wezel and Ruef[16, 17], as well as Sgourev and Van Lent Sgourev and van Lent [11], have recognized the

challenges inherent in the VOC dataset. Sgourev and Van Lent opted for a simpler approach, manually standardizing the places of origin for a random 100-record yearly sample and then applying a fuzzy coding algorithm to estimate the nationalities in the remaining records. However, they did not address reenlistment, assuming low rates based on Bruijn et al. [2]. Yet, Bruijn et al. actually observed frequent reenlistments among sailors in their analysis of a sample from the undigitized data.

In contrast, Wezel and Ruef undertook a more rigorous process. For their analysis of labor control mechanisms aboard ships of the VOC[16], they identified 561,454 seafarers with one or multiple employments records. Further, they write that they were able to ‘unambiguously’ assign the place of origin in 82% of the records to a region. However, their methods and data remain undisclosed, despite its potential value to other researchers.

We followed a similar approach to Wezel and Ruef in enhancing the VOC-opvarenden dataset: by disambiguating muster records (linking individuals to their respective records) and standardizing the places of origin. Additionally, we organized the ranks in the records and integrated wage data.

This article details how we processed the original dataset to come to the enriched version. Much of our process is replicable via the Jupyter notebooks we have published alongside the data files. Initially, we describe the original dataset, followed by a discussion on the applied enrichments. We then outline in Section 3.5 how we transformed the three original files into seven enriched ones. Lastly, Section 4 explores the reuse potential of our enriched dataset.

2 Original Data

The original dataset consists of three files.³ Since the dataset description is no longer available on the website of the Dutch National Archives⁴, we will describe these datasets in some detail.

2.1 Description

VOC-soldijboeken This dataset provides the origin of muster records. It comprises 3,248 entries detailing the source type, the associated ship's name, and the relevant VOC chamber (one of six semi-independent subsidiaries forming the VOC). These entries include ID numbers that link to record IDs in the VOC-opvarenden dataset, allowing for the association of each personnel record with its source. Additionally, voyage numbers from the Dutch-Asiatic Shipping database (see Footnote 1) are included.

Most entries (93%, or 3,020 records) in this dataset pertain to pay ledgers used by Company bookkeepers to track personnel departing from the Dutch Republic on specific ships. These ledgers correspond to 95% (737,467) of all muster records in the VOC-opvarenden dataset. The remaining entries cover 'verzoekboeken' (request books, documenting salary payment requests by VOC personnel for individuals in the Netherlands, available only for Amsterdam, Hoorn, and Zeeland chambers) and regiment books (detailing foreign soldiers hired by the VOC). The dataset includes 142 request books with 30,432 personnel records (4% of the total) and 86 regiment registers with 6,301 records (1% of the total).

VOC-opvarenden This is the main collection. It comprises 774,200 muster records from 1633-1794. Each record represents an individual enlisting on a VOC ship from one of the six chambers in the Netherlands in a certain year. Not included are personnel hired in Asia, including many Asian sailors, and enslaved individuals. Records detail the employee's origin, rank, service termination details, and reason for termination, with common reasons including completion of service, death, desertion, shipwreck, or becoming a free citizen in Asia.

The dataset indicates if the employee signed a transport or month letter. Transport letters, with a standard value of 150 guilders for sailors and higher amounts for senior ranks, were transferable salary claims, often used to clear pre-employment debts. The dataset represents this as a binary variable, without details on the beneficiary or the letter's value.

Month letters allowed employees to designate beneficiaries (often family members) to receive a portion of their salary, typically three months' wages annually. Transport and month letters were priority claims. A sailor with such letters would not receive wage payments until sufficient earnings were accrued to cover the yearly allotments to month letter beneficiaries and to settle any transport letter debts, in that order. If a sailor with outstanding transport or month letters died while in service, a bookkeeper calculated the salary earned up to the date of death. This amount, after deducting any debts to the Company, was allocated to the month letter beneficiaries and transport letter holders.

The dataset includes a binary representation for month letters, with beneficiary details in the third dataset, VOC-begunstigden. The dataset

³https://www.nationaalarchief.nl/sites/default/files/index_downloads/NL-HaNA_NT00444.zip.

⁴It is still available from internet archives, see e.g. <https://web.archive.org/web/20180203233056/http://www.gahetna.nl/collectie/index/nt00444/gebruik>

Chamber / source type	# records	Month letter		Transport letter	
		# records	% of records	# records	% of records
Amsterdam pay ledgers	394,260	35,137	8.9%	301,280	76.4%
Amsterdam request books	143	28	19.6%	135	94.4%
Delft pay ledgers	47,065	4,697	10.0%	33,661	71.5%
Enkhuizen pay ledgers	44,209	5,106	11.5%	35,973	81.4%
Hoorn pay ledgers	43,350	6,034	13.9%	34,953	80.6%
Hoorn request books	19	4	21.1%	18	94.7%
Rotterdam pay ledgers	45,580	5,324	11.7%	32,644	71.6%
Zeeland pay ledgers	163,003	8,956	5.5%	125,573	77.0%
Zeeland request books	30,270	6,174	20.4%	25,571	84.5%
Regiment books	6,301	0	0.0%	0	0.0%
Total	774,200	71,460	9.2%	589,808	76.2%

Table 1. Statistics on month and transport letters, sorted per VOC chamber and source type. Please note that the count of records with a month letter (71,460) is lower than the number of beneficiary records (76,337), due to the possibility of multiple beneficiaries associated with a single month letter.

records month letters signed in the Netherlands only, omitting those signed en route or in Asia. Table 1 in the paper provides a detailed breakdown of month and transport letters in the dataset.

The last piece of information that was taken from the original source documents reveals if a crew member switched ships during a stop at the Cape of Good Hope, a routine break for VOC ships en route between Europe and Asia since the mid-seventeenth century. If a crew member transferred ships during this outbound journey, a new entry was made in the receiving ship's pay ledger. This entry included the name of the previous ship and was usually closed again after arrival in Asia. The original pay ledger record remained open until the crew member's service ended.

Original pay ledger entries, accessible via links in most records, contain additional details not found in the database, like wage data and transport letter values. They also provide insights into an individual's career in Asia and wage evolution.

van Velzen [14] details the case of Andries Jurriaen Scholts from 'Luenenburg' (i.e. Lüneburg in Germany), hired on 14 July 1744 by the Amsterdam chamber as a soldier on the *Diemermeer*, with a starting wage of 9 guilders per month. Scholts signed a transport letter of 150 guilders to a person named 'J. Schroder'. Scholts remained in service for 21 years, entirely in Ceylon, where his wage eventually increased to 40 guilders per month. The ledger entry does not record the ranks which he held, however.

While some information from the original ledgers was excluded from the VOC-opvarenden database, new variables were added. These include metadata, German and English translations of ranks and contract termination reasons. If a crew member embarked on a homeward voyage, additional details on this voyage were included from the Dutch-Asiatic Shipping database. This enrichment provides comprehensive voyage data, such as departure and arrival dates and general remarks.

VOC-begunstigden This dataset, comprising 76,337 records, details the month letter beneficiary relationships with VOC employees. Each

The image shows a double-page spread of a handwritten ledger. The left page is titled 'In t Schip Dimermeek' and the right page is titled 'In t Schip van Amsterdam'. Both pages contain entries for 'Andries Jurriaen Scholts' from 'Luenenburg'. The entries are written in Dutch and include dates, amounts, and descriptions of transactions. The left page has a 'Somma' (Total) of 112 C. The right page has a 'Somma' of 112 C. The ledger is written in a cursive script typical of the 18th century.

Fig. 1. Example record of Andries Jurriaen Scholts from 'Luenenburg'. <http://proxy.handle.net/10648/cef964ad-a553-5c97-39e3-9aa43b1c81c6> and <http://proxy.handle.net/10648/354266c6-2546-7aea-6d27-7dad3cf2c7b5>

record lists the beneficiary's name, their relation to the employee (e.g. wife, mother, child; if the beneficiary was an institution, often an orphanage, this field reads 'instelling'), remarks (including institution names where applicable), and the corresponding VOC-opvarenden ID. A separate record is available for each named beneficiary. There are, for example, four records related to senior barber-surgeon Semein de Boer from Enkhuizen: one for each of his children.

2.2 Completeness

To assess the completeness of available personnel records, we utilized data from the Dutch-Asiatic Shipping database, which provides a comprehensive list of VOC voyages between the Dutch Republic and Asia. Table 2 presents statistical insights into source availability by decade.

Our analysis reveals that the records prior to 1680 are largely fragmentary. However, for the final two decades of the seventeenth century, pay ledgers are

available for nearly half of the outgoing voyages, supplemented by data from request books. Starting from the early eighteenth century until 1780, we have access to an almost complete collection of pay ledgers. In the latter years of the VOC, fewer pay books are available. This can be attributed not to source loss over time but to changes in the VOC's hiring policies: during this later period, the VOC engaged foreign regiments of soldiers (for which data are accessible from regiment books) and chartered foreign ships with their crews [14]. Although these chartered ship voyages are included in the Dutch-Asiatic Shipping database, they did not involve seafarers mustering with the VOC, leading to the absence of pay ledgers.

In summary, structural analyses of the VOC's seafaring workforce are best supported by data from the eighteenth century, which offer a more comprehensive dataset.

3 Data Enrichment

To adapt the original VOC maritime workforce data for large-scale analysis, we performed four enhancements: individual identification for career tracking, standardization and geolocation of places of origin, crew ranks structuring, and refining voyage data with improved dating.

3.1 Persons and Careers

Identifying individuals in the dataset involves linking muster records of the same person. This is complex due to the absence of an external validation source. Our approach is based on record analysis and logical inference to best link related records.

We describe the steps taken to cluster records likely associated with the same individual. This process involves normalizing names, applying a clustering algorithm, and evaluating the results. Our aim was to

⁵See <http://www.gerritbloothoof.nl/Publications/VREGELS.pdf>.

create high-quality clusters for statistical career analysis of VOC personnel, aiming to minimize false positives. (In other words, we did not try to find the highest number of possible matches between records, at the risk of sometimes linking records that in fact relate to two or more individuals, but to obtain a set of high-quality clusters.) Consequently, we used limited spelling normalization and strict record linkage rules. Our method leaves numerous 'ambiguous' records, potentially forming additional clusters for different analytical purposes.

3.1.1 Normalizing Person Names

Spelling variations in person names complicate record clustering. To address this, we developed a tailored set of rules for name rewriting, inspired by Gerrit Bloothoof's semi-phonetic Dutch name normalization.⁵ However, considering the international nature of the VOC workforce and varied linguistic knowledge of clerks, we limited our rule application. Our focus was on removing punctuation and common (but spelled variably) suffixes of patronymics and surnames, and normalizing interchangeable letter combinations (e.g. 'ph' and 'f').

3.1.2 Identifying Individuals

Based on several assumptions about when two records are likely to relate to the same person, we applied the following criteria to link more than one record to the same individual:

Person name The normalized person names in the records are identical or differ from each other by at most a Levenshtein distance (edit distance) of 1.

Place of origin The records contain the same unique standardized places of origin (see Section 3.2), or normalized places of origin that are identical or differ from each other by at most a Levenshtein distance (edit distance) of 1.

Decade	# outward voyages	# pay ledgers	# request books	# regiment books	% of voyages for which pay ledger is available
1600-1609	96	0	0	0	0.0%
1610-1619	112	0	0	0	0.0%
1620-1629	145	0	0	0	0.0%
1630-1639	142	7	0	0	4.9%
1640-1649	181	10	0	0	5.5%
1650-1659	205	0	0	0	0.0%
1660-1669	241	4	0	0	1.7%
1670-1679	225	5	29	0	2.2%
1680-1689	204	88	49	0	43.1%
1690-1699	233	108	46	0	46.4%
1700-1709	272	255	2	0	93.8%
1710-1719	318	307	2	0	96.5%
1720-1729	377	360	6	0	95.5%
1730-1739	381	373	2	0	97.9%
1740-1749	314	307	1	0	97.8%
1750-1759	292	287	1	0	98.3%
1760-1769	295	294	0	0	99.7%
1770-1779	293	292	0	0	99.7%
1780-1789	290	218	3	43	75.2%
1790-1794	133	102	1	42	76.7%
Total	4787	3017	142	85	

Table 2. Statistics on availability of sources per decade. The table does not include sources that could not be linked to a DAS voyage (3 pay ledgers and 1 regiment book).

No period overlap The voyages related to the records do not overlap in time. That is, the start date of the later voyage comes after the end date of the earlier voyage.

No long gaps The end date of the earlier voyage and start date of the later voyage are less than 7 years apart. (Although we can never be completely sure that two voyages were made by the same person, the probability decreases as the gap between successive voyages increases—with a long interval between voyages, we might, for example, confuse fathers and sons. The choice for the maximum interval of 7 years is based on

a review of batches of clusters where varying interval criteria were applied.)

Death is the end An employment record ending in death, shipwreck, murder or death sentence cannot be linked to a later record.

Using these criteria, we linked each individual record, except those records that describe crew members who boarded a ship at the Cape of Good Hope, to any other record (again, except Cape boarders) that could represent the same person. We thus turned the list of records into a graph of connected records. Once a record was linked to another, all linked name variants were used to link additional records.⁶

⁶This is to account for e.g. the following scenario. Two records, r_1 and r_2 are linked; the Levenshtein distance between their person names is 1 and the voyage end date of r_1 comes 1 year before the voyage start date of r_2 . Now consider a non-linked record r_u that has a person name with a Levenshtein distance of 1 with record r_1 and 2 with record r_2 , and a start date 8 years after the end date of r_1 and 2 years after r_2 . We consider it likely that r_u describes the same person as r_1 and r_2 . We link r_u to the cluster because r_1 and r_2 together fit the criteria for linking to r_u .

Additionally, we integrated records of crew members who boarded at the Cape of Good Hope, using similar criteria for personal name and birthplace similarity, and ensuring ship name consistency in the original and Cape boarder records.

The linking process resulted in two record categories:

Disambiguated Any cluster of records that does not violate any of our criteria for assuming that records represent the same person, is considered to represent an individual who re-enlisted one or multiple times. Any record that could not be linked to another record according to the criteria for matching person names and places of origin described above, is considered to represent an individual who went on a single voyage for the VOC.

Ambiguous Any record linked to another based on the criteria for matching person names and places of origin, but violating at least one of the other criteria.

3.1.3 Statistics

Our process yielded 547,132 disambiguated records, corresponding to 460,452 individuals (Table 3). The distribution of voyages per individual is illustrated in Figure 2, with a majority undertaking a single voyage. The logarithmic scale of the right chart in Figure 2 indicates a power-law distribution, suggesting the absence of data artifacts and reliability of our clustering. (A sharp drop or a flattening of a logarithmic curve is usually a signal of some artefact. For example, if this curve were to drop sharply after, say, six voyages, this could be a signal that our data do not cover a sufficiently long period, so that first or last voyages of sailors who make more than six voyages are missed. However, this is not the case.) We interpret this as a signal that our clustering of records by person is reliable. Nonetheless, a thorough evaluation process, detailed in the following section, was undertaken to further affirm the quality of our clustering.

Aspect		Number
Records	Total	774,200
	Disambiguated	547,132
	Ambiguous	227,068
Person names	Raw	554,810
	Normalized	493,653
Persons		460,452
Place names	Raw	152,315
	Normalized	143,101
Countries		99
Ranks		197
Outward voyages		3,268

Table 3. Descriptive statistics of the enriched dataset. Note that the number of outward voyages is higher than the number of pay ledgers in the original VOC-soldijboeken dataset (3,020). Despite the absence of pay ledgers for certain outbound voyages, details about some crew members were retrieved through records from subsequent ships they boarded at the Cape.

3.1.4 Evaluation

Evaluating our clustering and individual identification in the dataset is challenging due to limited information. Despite identical names, origins, and ranks in records with closely timed journeys, we cannot definitively confirm these records refer to the same individual. However, we do have a key piece of external information for evaluation: month letter beneficiaries.

When clustered individuals consistently nominate the same beneficiary across multiple voyages, it strengthens the likelihood that the cluster represents a single person, although exceptions may occur due to common names in that era. (There may for example have been multiple men in eighteenth-century Amsterdam with the same name, whose mothers also shared names.)

As detailed in Section 2.1, each of the 76,337 beneficiary records in the VOC-begunstigden dataset links to a muster record. Data shows that employees mostly nominated wives (53.4%) and mothers (25.8%) as beneficiaries, with a higher wife nomination rate (61.4%) in multi-journey clusters (Table 4).

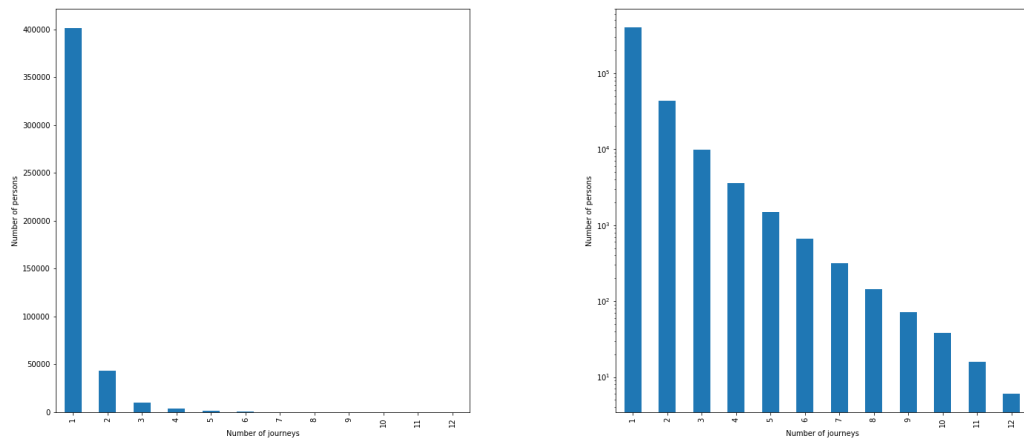


Fig. 2. The distribution of the number of times that (disambiguated) individuals enlisted with the VOC. The Y axis has a normal scale in the chart on the left and a logarithmic one on the right.

Relation	All beneficiaries		Cluster beneficiaries	
	# records	fraction	# records	fraction
Wife	40,758	0.534	9,183	0.614
Mother	19,684	0.258	3,594	0.240
Father	5393	0.071	937	0.063
Child	3520	0.046	552	0.037
Children	559	0.007	87	0.006
Sister	821	0.011	24	0.002
Brother	380	0.005	11	0.001
Institution	627	0.008	48	0.003
Other	618	0.008	27	0.002
Unknown	3977	0.052	488	0.033
Total	76,337	1.000	14,951	1.000

Table 4. Distribution of beneficiary relation types in the whole VOC-opvarenden dataset (columns 2 and 3) and related to the multi-voyage clusters (columns 4 and 5).

Relations	# records	fraction
No record with beneficiary relation	35,864	0.80
Single record with beneficiary relation	5,261	0.12
Multiple records with beneficiary relation; no overlapping relation	614	0.01
Multiple records with beneficiary relation; overlapping relation	3,007	0.07
Total clusters	44,746	1.00

Table 5. The distribution of month letter beneficiary relations registered in multi-voyage clusters.

Beneficiary data aids in verifying clustering if at least two muster records in a cluster record the same beneficiary. However, most clusters (92%) have no or just one beneficiary recorded per journey (Table 5). Of the 3,621 clusters with at least two beneficiary-linked records, 614 have varied beneficiary types, offering no evaluative insight. Only 3,007 clusters with recurring beneficiaries are useful for validation.

Relation	# clusters	fraction
Wife	2,376	0.76
Mother	502	0.16
Father	107	0.03
Child	118	0.04
Children	8	0.00
Brother	1	0.00
Sister	1	0.00
Unknown	4	0.00
Other	1	0.00
Total	3,118	1.00

Table 6. Distribution of month letter beneficiary relations for multi-voyage clusters with the same relation type for multiple voyages.

Table 6 displays recurring beneficiary relations within these 3,007 clusters. More than one recurring relation type per cluster is possible, like multiple journeys with wife or parents as beneficiaries.

We categorize clusters with recurring beneficiary relations for validation as follows:

- (1) Complete normalized name⁷ overlap: beneficiaries with the same relation all have the same name, indicating high cluster accuracy.
- (2) No normalized name overlap: beneficiaries with the same relation all have different names. This is possible if the beneficiary is the employee's wife or child, as the employee may have remarried and may have had multiple children. But it could also be a sign that the cluster is incorrect, especially if the beneficiary relation is mother or father.

⁷We applied the same name normalization that we described in Section 3.1.1.

- (3) Ordered normalized name overlap: some beneficiaries with the same relation share names and they are grouped in temporal order (e.g. A, A, B or A, B, B, or A, B, C, C). Again, possible for wife/child, not for mother/father.
- (4) Unordered normalized name overlap: non-sequential name sharing for the same relation, strongest indication of incorrect clustering except for child beneficiaries.

Table 7 reveals that beneficiary relations support our clustering in 92% of cases. However, upon examining clusters with non-matching names for the same relationships (the remaining 8% of clusters), we found that many beneficiary names within a cluster are phonetically very similar but orthographically different. In some cases, the transcribers seem to have swapped the patronymic name and last name of a person, or have entered the patronymic name as the last name with no proper last name given. In many cases, the registered first names are variants of each other, as in *Maria* and *Marietje*, *Elizabeth* and *Betje* or *Catharina* and *Trientje*, or one record has only initials while the other has the full name.

Analysis of 100 clusters with differing spouse names but no conflict in order showed 52 with phonetically similar or variant names. Among 79 clusters with unordered spouse names, suggesting incorrect clustering, 46 had potentially matching names. For parent name mismatches, 83 out of 161 had plausible name similarities.

In summary, over half of the sampled clusters with non-matching names can be attributed to name variations, transcription errors, and use of initials or full names. This suggests a lower rate of potentially invalid clusters and a higher consistency in beneficiary naming than initially indicated by Table 7.

Determining precise clustering quality remains difficult, but this validation with independent data reinforces our confidence in the overall accuracy of the

Status	# clusters	fraction
Complete normalized name overlap	2,103	0.70
Ordered normalized name overlap wife	664	0.22
Unordered normalized name overlap wife	76	0.03
Ordered normalized name overlap other	0	0.00
Unordered normalized name overlap other	1	0.00
No normalized name overlap father or mother	146	0.05
Total	3,007	1.00

Table 7. Distribution of recurring beneficiary relations in multi-voyage clusters.

clusters. We believe that more precise clustering would not significantly alter outcomes in large-scale analyses of the VOC-opvarenden dataset and the career trajectories of individuals within the VOC.

3.2 Places

This section outlines our process for standardizing and geolocating the origins of VOC crew members. We successfully standardized 44,152 of the original 152,315 unique place of origin attestations (about 30%) to 8,591 modern toponyms with geographical coordinates. This allows us to locate the origins mentioned in over 82% of the muster records.

3.2.1 Place Name Standardization Process

Re-use of previous work Our approach began by capitalizing on the efforts of past projects to standardize and geocode toponyms in the VOC-opvarenden dataset. We found the Dutch Ships and Sailors (DSS) data of insufficient quality to build upon, but did adopt the ErfGeo project's standardized modern toponyms of 4,194 original attestations of Dutch places as our foundation.⁸ Additionally, we utilized Clé Lesger's database, which focused on Dutch historical toponyms from citizenship registers of early modern Dutch cities (so-called 'Poorterboeken').⁹

We then incorporated some data from the Sound Toll Registers online project, which offered standardized names for early modern attestations of port towns, especially in Scandinavia and the Baltic.¹⁰ Finally, we integrated data from Ton van Velzen, who led the digitization of the VOC pay ledgers and later worked on standardizing toponyms in the original data. Although his work lacked URIs for clear identification, it provided standardized places for 2,100 original attestations, which proved valuable in our overall standardization process.

Bigram clustering Using the bigram clustering function of OpenRefine¹¹, we matched as yet unstandardized toponym attestations to attestations for which we had a standardized equivalent from one of the datasets described above.

Manual standardization Our data curation team undertook a manual process to review and augment the data obtained from earlier steps. Faced with numerous unstandardized and ambiguous toponym attestations, our goal was not to standardize every attestation but to accurately identify as many VOC crew members' places of origin as possible. We prioritized toponyms based on their frequency of occurrence, addressing the most common ones first.

⁸See <https://linkeddata.cultureelerfgoed.nl/rce/erfgeo> and <https://github.com/erfgoed-en-locatie/data/tree/master/voc-opvarenden>.

⁹The data (2,784 records) are available from the ErfGeo GitHub repository: <https://github.com/erfgoed-en-locatie/data/tree/master/poorterboeken>.

¹⁰See <http://www.soundtoll.nl/>.

¹¹<https://openrefine.org/>

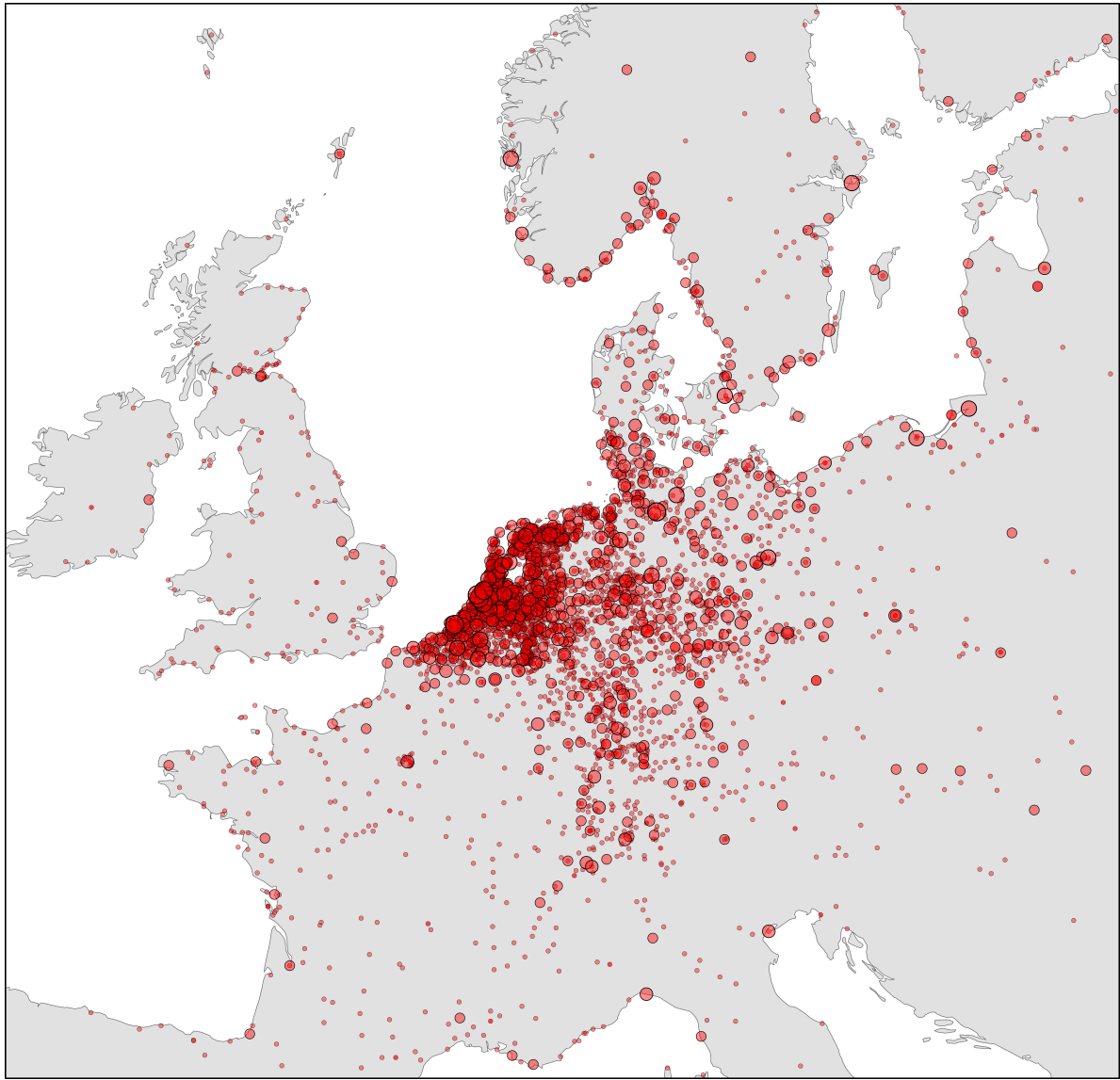


Fig. 3. Map showing the places of origin of individuals who enlisted with the VOC in the eighteenth century.

Several key resources significantly aided our efforts. Notably, Simon Hart's compilation of nearly 60,000 place of origin attestations for seventeenth and eighteenth-century Amsterdam migrants provided valuable parallels to our VOC-opvarenden dataset.¹² Similarly, Sølvi Sogner's (unpublished) collection of about 11,500 Scandinavian place name attestations offered essential

insights. Google Maps also helped us locate and standardize historical place name attestations. Its algorithms handle spelling variations and historical and/or foreign-language spelling of place names very well.

When a single clear modern place name candidate for a historical toponym emerged using the aforementioned resources, curators recorded the

¹²Hart's data are available from https://assets.amsterdam.nl/publish/pages/753563/herkomstonderzoek_2.pdf.

modern toponym and the country's ISO 3166-1 alpha 2 code. They were sometimes helped by additional information that gave a clear indication which place was meant by it. 'Soest in brandenburg' could e.g. be standardized as Soest in present-day North Rhine-Westphalia. If there were multiple plausible candidates, they did nothing and proceeded to the next attestation. If there were multiple candidates, with one being clearly more plausible than the others, they entered the modern toponym, the country code, and, as a final unambiguous identification, the GeoNames URI of the place. They based their assessment of plausibility on two criteria: 1) size—if one place clearly has more inhabitants than the other(s), and it is likely that this was also the case in the eighteenth century, we argue that it is plausible that the VOC employee came from the large place; 2) location—if one place is located in a region where many VOC employees came from (the Netherlands, coastal and central Germany, the Baltic region, and coastal regions of Scandinavia), and the other(s) not, we argue that it is plausible that the VOC employee came from the place located in the core area. Once a likely place of origin was identified, curators also standardized similar toponyms from the list.

GeoNames and Wikidata Resolution

We semi-automatically resolved the toponym-country code combinations from the next step to GeoNames, retrieving a maximum of three populated places from the GeoNames database¹³, and then manually reviewing the results with more than one GeoNames candidate. We only finalized the standardization if we considered just one result to be plausible. For historical regions, we manually linked to Wikidata.¹⁴ Finally, we

retrieved latitude and longitude values from GeoNames and Wikidata using the respective URIs. If the Wikidata record did not provide geographical coordinates, we took the latitude and longitude values of the largest town within the region. After this step, we had standardized names with geographical coordinates for over 37,000 original toponyms.

GIS map matching In collaboration with Maarten Bosker and Dinand Webbink from Erasmus University Rotterdam, we further enhanced our dataset using a collection of 37 historical GIS maps. This approach involved fuzzy matching the original toponyms with placesnames found on the maps and reconciling these matches to GeoNames. The methodology added standardized places for an additional 7,053 original toponym attestations.

Our methodology comprised three main steps. First, we cleaned the original toponym attestations by removing punctuation and additional details (e.g., 'in brandenburg'). We then fuzzy matched these to the GIS collections, selecting the best match for each. In cases of multiple matches with identical similarity scores, we chose the match in or nearest to the Netherlands. We eliminated single-character toponyms and retained only exact matches or those with additional information (like 'in brandenburg') for toponyms with 2-4 characters. For toponyms with five or more characters, we kept matches with a similarity score of at least 0.78, as scores below this threshold yielded less than 5% plausible matches.

¹³The API call we applied was <http://api.geonames.org/searchJSON?q=TOPONYM&country=COUNTRYCODE&maxRows=3&fclass=p&fuzzy=0.99&username=USERNAME>, where `fclass=p` limits the search results to populated places. Adding `fuzzy=0.99` to the API call turned out to somehow yield more precise results than when this parameter was left out.

¹⁴<https://www.wikidata.org/>

In the second step, we reconciled matches with GeoNames, using it also for ambiguity checks.¹⁵ To assess match ambiguity, we used GeoNames to identify similar names in regions predominantly associated with VOC employees.¹⁶ Finally, we accepted all standardized toponyms that were both reconciled to GeoNames and had no other plausible GeoNames matches. We dismissed ambiguous matches identified in multiple locations by GeoNames. Additionally, we conducted a clerical review of all GIS-file-matched places not reconciled to GeoNames and those with supplementary information.

Assignment of places to regions The final step was assigning standardized places to specific regions based on language, culture, maritime orientation, and proximity to the Netherlands. The regions are depicted in Figure 4 and can broadly be defined as follows:

- A** Dutch Republic and Dutch-speaking area of Southern Netherlands;
- B** Low German speaking area;
- C** Scandinavia;
- D** German interior lands;
- E** British Isles;
- F** France;
- G** Iberian Peninsula;
- H** modern-day Italy and Corsica;
- I** Eastern and Southeastern Europe.

The chart on the right in Figure 4 clearly shows that the majority of seafarers hired by the VOC in the eighteenth century came from the Dutch Republic and Dutch-speaking area of the Southern Netherlands (region A). Most ‘foreigners’

on board came from regions B, C and D. The supply of French crew members (region F) is small but not insignificant. It should be noted, however, that their presence is highly skewed towards the end of the eighteenth century, when the Netherlands increasingly came under French influence. The other regions supplied relatively very few employees. The same goes for the rest of the world: the dataset contains muster records of people from North America, the Caribbean, South America, Africa and Asia, but their numbers are very low.

3.2.2 Discussion of Results

The standardization of the place of origin attestations in the original VOC-opvarenden dataset resulted in 82% coverage of the 152,315 unique toponym attestations from the muster records. However, it’s crucial to critically examine the results for potential bias and standardization quality.

Possible bias As Figure 4 indicates, most muster records pertain to the Dutch Republic and Dutch-speaking Southern Netherlands, reflecting historical reality and aligning with previous research [2, 3]. However, a potential bias towards Dutch-speaking European places might exist for two reasons. First, VOC clerks were likely more familiar with Dutch places and had easier communication with prospective employees from these regions, probably resulting in less spelling variation. The transcription of toponyms was also done by Dutch people, again probably leading to fewer errors and lower spelling variation for Dutch toponyms. Second, the standardization process, which used some resources focused

¹⁵The standardized places from the GIS files came with geographical coordinates. To geographically focus the GeoNames reconciliation, we drew a rectangle of 0.4 degrees long and wide (i.e. around 1,500 square kilometers; the exact size being dependent on the north-south position) around the geographical location of the standardized place and retrieved matching entities within this rectangle. The API call we applied was http://api.geonames.org/searchJSON?q=TOPONYM&south=SOUTHERN_BOUNDARY_RECTANGLE&north=NORTHERN_BOUNDARY&east=EASTERN_BOUNDARY&west=WESTERN_BOUNDARY&fclass=P&maxRows=1&fuzzy=0.99&username=USERNAME.

¹⁶The API call we applied was <http://api.geonames.org/searchJSON?q=TOPONYM&fclass=P&fuzzy=0.99&country=NL&country=BE&country=DE&country=FR&country=NO&country=SE&country=FI&country=LU&country=GB&country=PL&country=LT&country=EE&country=LV&country=AT&country=CH&country=CZ&username=USERNAME>.

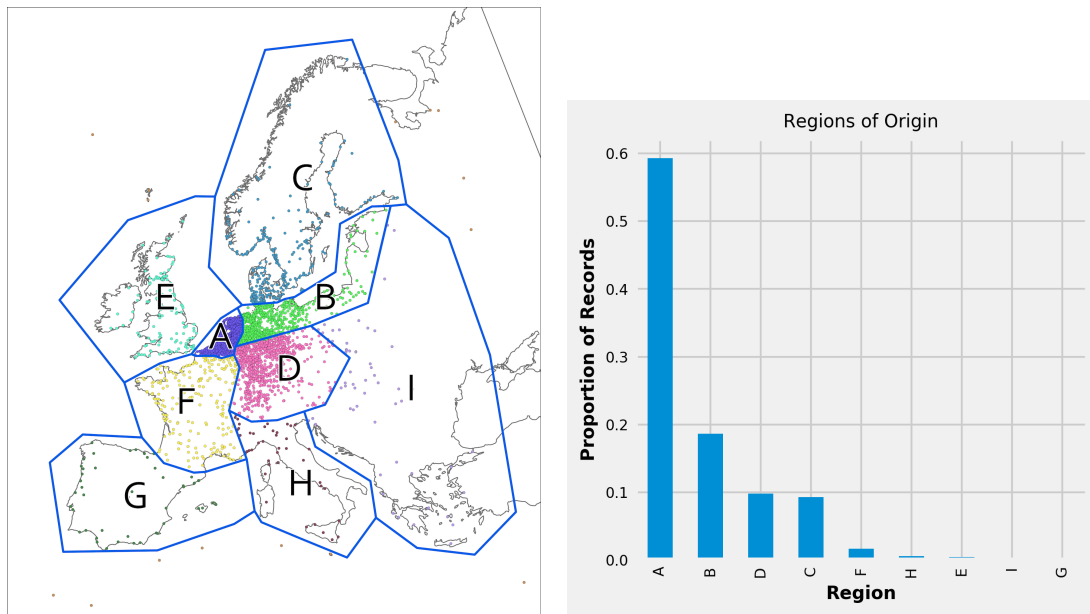


Fig. 4. Map showing the region categorization (*left*) and distribution of disambiguated records over the regions.

on the Netherlands and involved Dutch data curators, might have favored Dutch places. Although the semi-automated linkage with GIS collections might have mitigated this bias, non-Dutch places could be over-represented in the unstandardized toponyms.

Quality of standardization Standardizing toponyms from an early modern source with limited context is imprecise. The process, partly based on assumptions and human judgment, likely produced incorrect matches. Despite these limitations, we believe the current set of standardized places accurately represents the origins of VOC crew members, making it valuable for large-scale analyses.

3.3 Ranks and Wages

The original dataset includes 197 unique Dutch-language ranks, with clear ones like ‘matroos’ (sailor) and ‘soldaat’ (soldier) accounting for 395,133 records. However, some ranks are ambiguous due to a change in meaning in 1784, when the VOC tried to attract

more skilled personnel by applying higher rank designations to lower ranks[5], and inconsistencies in digitization.

Our adaptations for career analysis and clarity are as follows:

Complete and improve English translation

We refined the English translations of certain ranks from the original dataset.

Create new ranks We introduced new ranks with the suffix ‘after 1783’ for those whose meanings changed in 1784.

Add wage sample To supplement the lack of wage data in the original VOC-opvarenden dataset, we gathered starting salary information for a sample of 5,335 records across various ranks, geographical origins and time periods.

Correct transcription errors We rectified common transcription mistakes, particularly in the ranks ‘opperscheepstimmerman’ (senior ship carpenter) and ‘opperstuurman’ (first mate), totaling 725 corrections.

Assign parent ranks Similar ranks were grouped under 107 parent ranks, using descriptions from the original dataset, our wage data, and the VOC glossary.¹⁷

Assign categories Ranks and parent ranks were classified into six categories: Trade, Sea, Ship, Medical, Military, and Other, based on their responsibilities. Trade contains the ranks of the personnel responsible for trading goods in Asia. Sea contains all maritime ranks: the captain, maritime officers and sailors. Ship is a collection of ranks charged with maintenance, order on board and food supply. Medical covers the barber-surgeon and his helpers. Military contains the soldiers and their officers. Finally, all ranks that do not fit in any of these five categories, such as clergyman and windmill maker, were grouped under Other.

Assign HISCO codes Each rank was assigned a code from the historical international classification of occupations (HISCO) to indicate hierarchy and work type.¹⁸

Table 8 summarizes our work on ranks. It shows the most frequently occurring parent ranks, hierarchically organized within their categories, and data from our wage sample. The high wage variance among captains and some other ranks is attributed to skill premiums and experience, with some variance in (junior) ship carpenters also due to inconsistencies in the standardization of these ranks during digitization.

3.4 Voyages and Dates

We undertook a cleaning process for the dates, correcting anomalies (like 1700-06-31) and removing incorrect ones. We also filled in incomplete contract end dates using the DAS database, assuming that a contract ended when the return voyage arrived in the Netherlands. This resulted in an enhancement of 218,405 contract end dates, about 30% of the total. For calculation

purposes, we standardized incomplete dates by changing '00' values in months and days to '01'. Additionally, we refined the DAS voyage numbers, manually checking and fixing typos and aligning them with more practical DAS voyage IDs.

3.5 Enriched Files

We publish the enriched data in eight files, respectively describing the sources, the seafarers and their employment contracts, their names, the places where they came from (two files), the ranks they held, the voyages on which they served, and the beneficiaries to whom they assigned their month letters. In this Section we briefly describe how we derived the eight files from the original three in the VOC-opvarenden dataset. We predominantly utilized Python for data transformations, with accompanying Jupyter notebooks providing insights into these processes. Additionally, Appendix A.1 offers brief explanations of all variables in these files, while Appendix A.2 illustrates their interconnections.

voc_sources.csv This file, with 3,246 entries, details the sources of the VOC crew members' muster records, mainly derived from the VOC-soldijboeken file. Transformations (see `create_voc_sources.ipynb`) include adding English column names, correcting DAS database references, replacing DAS voyage IDs with more user-friendly IDs, specifying source types, and removing redundant IDs of muster records.

voc_persons_contracts.csv This is the main file in our collection. It describes the muster records ($n = 774,200$) and indicates which records we managed to disambiguate ($n = 547,132$), resulting in data on individuals ($n = 460,452$). This file is largely based on the VOC-opvarenden file in the original dataset. The notebook `create_voc_persons_contracts.ipynb` shows the transformations that we performed:

¹⁷See <http://resources.huygens.knaw.nl/vocglossarium/zoekvoc>

¹⁸See <https://historyofwork.iisg.nl/index.php>

Category	Parent rank	Median wage	Average wage	Standard deviation	<i>n</i> wage sample	# records in dataset
TRADE	senior merchant	135	115.09	41.01	44	146
TRADE	merchant	40	41.19	5.50	42	1,322
TRADE	merchant's assistant	24	40.95	28.06	55	1,629
SEA	captain	72	73.36	11.85	315	3,134
SEA	first mate	48	47.92	2.06	294	3,239
SEA	second mate	32	32.18	2.78	234	3,726
SEA	third mate	26	26.05	0.70	202	5,074
SEA	boatswain	22	19.92	3.84	134	6,909
SEA	junior boatswain	14	14.01	0.44	67	5,801
SEA	quartermaster	14	13.98	0.65	60	10,845
SEA	sailor	11	10.56	1.35	540	274,455
SEA	junior sailor	7	6.68	1.16	244	96,036
SHIP	ship carpenter	48	45.37	4.19	144	2,412
SHIP	junior ship carpenter	28	28.20	7.93	451	16,488
SHIP	senior sail maker	20	20.46	2.55	48	2,683
SHIP	sail maker	16	16.67	2.94	27	814
SHIP	junior sail maker	14	13.75	1.83	32	2,902
SHIP	weapons master	22	21.92	0.52	74	3,228
SHIP	junior weapons master	14	14.09	1.53	75	7,074
SHIP	master-at-arms	13	12.57	0.50	35	2,800
SHIP	cook	20	20.07	0.86	92	3,263
SHIP	junior cook	14	14.07	0.47	41	3,098
SHIP	water maker	18	17.61	1.50	31	215
SHIP	junior water maker	12	12.00	0.00	9	111
SHIP	steward	20	19.50	1.71	60	3,235
SHIP	junior steward	14	11.61	2.86	49	3,177
SHIP	senior cooper	16	16.69	1.22	91	2,775
SHIP	cooper	14	14.51	2.09	55	827
SHIP	junior cooper	14	13.82	0.80	98	3,902
MEDICAL	senior barber-surgeon	36	36.74	2.51	113	3,173
MEDICAL	barber-surgeon	26	24.99	3.64	162	4,092
MEDICAL	junior barber-surgeon	14	14.15	1.64	104	2,768
MILITARY	military captain	80	81.56	17.31	9	120
MILITARY	lieutenant	50	47.58	8.47	38	193
MILITARY	ensign	40	40.00	0.00	11	91
MILITARY	bombardier	24	24.67	5.16	6	72
MILITARY	sergeant	20	20.00	0.00	61	3,124
MILITARY	corporal	14	14.00	0.00	46	6,227
MILITARY	lance-corporal	12	12.79	1.03	86	6,038
MILITARY	soldier	9	9.21	0.47	197	241,102
MILITARY	junior soldier	7	7.00	0.00	23	1,844
MILITARY	weapon maker	14	13.97	0.80	87	2,230
MILITARY	junior weapon maker	9	9.00	0.00	2	92
Total					4,588	742,486

Table 8. Descriptive statistics of wage sample. In this table, the parent ranks are hierarchically organized per category. Note that the SHIP and MILITARY categories have one or more sub-hierarchies. Note also that the OTHER category and certain less frequently occurring parent ranks from the five main categories (together representing 31,714 records or 4% of the total) were omitted from this table.

we added cluster data that link records to individuals (please note, the clustering itself was performed outside this notebook; this process was described at length in Section 3.1), added references to standardized places and ranks, calculated contract lengths, and removed superfluous columns.

voc_names.csv To decrease the file size of the previous file, we moved the names (both those transcribed from the muster records and their normalized variants) to this separate file.

voc_places.csv To further decrease the file size of the `voc_persons_contracts.csv` file, we moved the unique attestations of places of origins ($n = 152,315$) and their normalized variants to this separate file.

voc_places_standardized.csv Lists standardized place names as detailed in Section 3.2.

voc_ranks.csv Includes the 200 unique ranks from the VOC-opvarenden file in the original dataset and their classification into 107 parent ranks (see Section 3.3).

voc_voyages.csv Provides data on 8,189 ship voyages crewed by the seafarers described in `voc_persons_contracts.csv`.

voc_beneficiaries.csv An enhanced version of the original VOC-begunstigden file, with translated column names and normalized relation information.

4 Re-use Potential, Future Work, and Corrections

Due to its considerable size, the dataset we present in this paper offers unparalleled opportunities for extensive, large-scale statistical analyses. Currently, two papers utilizing this dataset are in preparation; one delves into career opportunities within the VOC, while the other examines the role of networks in recruiting international workers. Yet, the research potential of

this dataset extends beyond these topics. It is ripe for sociological analyses, akin to the studies by Sgourev and van Lent [11], Wezel and Ruef [16, 17], and has great potential for network studies, potentially revealing insights into recruitment and social networks within the VOC. When merged with data on economic and social factors from various European regions, the dataset could facilitate in-depth studies on migration patterns, among other topics.

The GLOBALISE project¹⁹ aims to incorporate this dataset into its reference datasets on individuals. By the end of 2026, the data will be accessible as Linked Open Data, facilitating integration with other datasets. Furthermore, GLOBALISE may enrich our dataset with information from the VOC's 'Land- en Zeemonsterrollen', annual records of the Company's workforce in Asia. This addition promises a more detailed understanding of the lives and careers of the VOC-opvarenden in Asia.

Additionally, GLOBALISE's source corpus, containing nearly 5 million scans of records, letters, and documents related to the VOC in Asia, offers qualitative data on individuals in our dataset. This rich corpus is particularly valuable for genealogists, who already frequent the original VOC-opvarenden dataset, providing them with extensive resources to trace their ancestors' lives and careers in Asia.

The dataset also appeals to audiences beyond academic and genealogical researchers. The 2021 VOC Data Experience²⁰, an augmented reality-based museum installation, draws from this enriched dataset to offer immersive visualizations that explore the complex and often challenging history of the VOC.

However, it is crucial to acknowledge the dataset's limitations, particularly regarding potential errors. The automatic clustering of records related to individuals is not infallible, occasionally merging records of different individuals or splitting records of the same individual. In addition, the standardization of places

¹⁹<https://globalise.huygens.knaw.nl/>

²⁰<https://vocdataexperience.nl/>

of origin remains incomplete, posing challenges for geographic analysis. The scarcity of contextual information also contributes to inherent ambiguities. Researchers should be mindful of these limitations when utilizing the dataset.

While we do not anticipate major enhancements to the dataset, we encourage the academic community and the public to contribute to its accuracy by submitting corrections or additions, especially regarding places of origin. If you have over 100 corrections or additions, please reach out to the first author (contact details provided on the first page).

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A Appendix

A.1 Description of files and variables

The enriched data set comes in seven files. The following tables describe the variables included in the files that respectively describe the sources (9), the persons and their contracts (10), the places of origin of the VOC employees (12 and 13), the ranks held by them (14), the voyages on which they served (15), and the beneficiaries to whom they assigned their month letters (16).

Column name	Provenance	Description
source_id	NA	record id; each record describes a physical archival document that contains data on the crew of a particular voyage of a VOC ship
ship_name	NA	name of ship for which the source document provides data on the crew of a particular voyage
chamber	NA & Huygens	VOC chamber that managed the ship voyage and its crew
remarks	NA	general remarks on record (in Dutch)
archival_reference	NA	reference to inventory number in VOC archives at NA
uid	NA	unique record id in database of Dutch National Archives
source_type	NA & Huygens	specification of source type (pay ledger, request book or regiment book)
das_voyage_id	NA & Huygens	unique id of voyage, refers to file VOC_voyages.csv

Table 9. Description of variables included in file `voc_sources.csv`

Column name	Provenance	Description
vocop_id	NA	record id; each record describes a muster record of a VOC employee that was recorded in the pay ledger of a particular voyage of a VOC ship
full_name	NA	full name of employee
place_of_origin	NA	place of origin of employee
place_id	Huygens	id of place of origin, refers to file VOC_places.csv
disambiguated_person	Huygens	1 if record belongs to a disambiguated person, else 0
person_cluster_id	Huygens	id for cluster of records belonging to a disambiguated person
person_cluster_row	Huygens	ordinal rank of the voyage in a disambiguated person's career ('3' would e.g. indicate the third contract within a cluster)
rank	NA / Huygens	rank designation in original data set, with some alterations by Huygens (correction of wrong transcription; distinction ranks that changed after 1783)
rank_corrected	Huygens	1 if the rank stated in the original data set has been altered (correction of wrong transcription; distinction ranks that changed after 1783), else 0
rank_id	Huygens	unique id of rank, refers to file <code>voc_ranks.csv</code>
debt_letter	NA / Huygens	1 if employee signed a debt letter, else 0
month_letter	NA / Huygens	1 if employee signed a month letter, else 0
date_begin_contract	NA	start date of employment contract

Continued on next page

Table 10 continued from previous page

Column name	Provenance	Description
date_end_contract	NA / Huygens	end date of contract; incomplete dates in original data set were supplemented with arrival dates of return ships from DAS and/or completed with ‘-01’ or ‘-01-01’ to facilitate calculations with the dates
contract_length	Huygens	length of contract (in days)
reason_end_contract	NA	reason why the employment contract ended
could_muster_again	Huygens	1 if employee could muster again on a subsequent voyage, given the reason why the current employment contract ended, else 0
location_end_contract	NA	location where the employment contract ended; when an employee returned to the Netherlands and signed off, the name of the return ship is given
outward_voyage_id	Huygens	DAS voyage id of outward voyage, refers to file voc_voyages.csv
changed_ship_at_cape	NA / Huygens	1 if employee changed ship at the Cape of Good Hope, else 0
changed_ship_at_cape_voyage_id	NA / Huygens	DAS voyage id of outward voyage that employee joined from the Cape of Good Hope, refers to file voc_voyages.csv
return_voyage_id	NA / Huygens	DAS voyage id of return voyage, refers to file voc_voyages.csv
remark	NA	general remark on record (in Dutch)
source_id	NA	id of source record from which muster record originates, refers to file voc_sources.csv
source_reference	NA	reference to finding place of record in the paper VOC archives (inventory number and page number)
uid	NA	unique record id in database of Dutch National Archives
scan_permalink	NA	permalink to scan of physical muster record

Table 10. Description of variables included in file voc_persons_contracts.csv

Column name	Provenance	Description
vocop_id	Huygens	muster record id from which name originates, refers to file voc_persons_contracts.csv
first_name_original	NA	first name of employee
patronymic_original	NA	patronymic of employee
family_name_prefix_original	NA	family name prefix of employee
family_name_original	NA	family name of employee
full_name_normalized	Huygens	normalized full name of employee
first_name_normalized	Huygens	normalized first name of employee
patronymic_normalized	Huygens	normalized patronymic of employee
family_name_prefix_normalized	Huygens	normalized family name prefix of employee
family_name_normalized	Huygens	normalized family name of employee

Table 11. Description of variables included in file voc_names.csv

Column name	Provenance	Description
place_id	Huygens	record id; each record describes a place of origin mentioned in the VOC muster records
place_original	NA	toponym attestation in original data set
place_normalized	Huygens	normalized toponym attestation
place_standardized_id	Huygens	unique id of standardized place, refers to file voc_places_standardized.csv; left blank if place has not been standardized
provenance	Huygens	code specifying the provenance of the standardization (1: ErfGeo VOC-opvarenden, 2: ErfGeo Poorterboeken, 3: Huygens, 4: Sound Toll, 5: NA / Ton van Velzen, 6: Huygens-Erasmus)

Table 12. Description of variables included in file voc_places.csv

Column name	Provenance	Description
place_standardized_id	Huygens	record id; each record describes a standardized place of origin of a VOC employee
place_standardized	Huygens	standardized placename
country_code	Huygens	ISO 3166-1 alpha 2 code of country in which standardized place is presently located
place_standardized_cc	Huygens	concatenation of standardized placename and country code
authority_file_uri	Huygens	GeoNames or Wikidata URI of standardized place (Wikidata used when GeoNames unavailable, e.g. for historical regions)
latitude	Huygens	latitude of place of origin, as provided by authority file
longitude	Huygens	longitude of place of origin, as provided by authority file
region	Huygens	region in which standardized place is located (A: Dutch Republic and Dutch-speaking area of Southern Netherlands, B: Low German-speaking area, C: Scandinavia, D: German interior lands, E: British Isles, F: France, G: Iberian Peninsula, H: Italy and Corsica, I: Eastern and Southeastern Europe)

Table 13. Description of variables included in file voc_places_standardized.csv

Column name	Provenance	Description
rank_id	Huygens	record id; each record describes a rank on board of a VOC ship
rank	NA / Huygens	rank designation
parent_rank	Huygens	parent rank, grouping together one or more similar ranks
category	Huygens	rank categorization (trade, sea, ship, medical, military or other)
subcategory	Huygens	further rank categorization, e.g. distinguishing between commissioned and non-commissioned officers
hisco	Huygens	HISCO code for rank (see https://historyofwork.iisg.nl/)
hico_uri	Huygens	URI for HISCO code
rank_nl	NA / Huygens	rank designation in Dutch
rank_description_nl	NA	description of rank in Dutch
rank_description_eng	NA	description of rank in English
median_wage	Huygens	median wage of rank, calculated from wage sample

Table 14. Description of variables included in file voc_ranks.csv

Column name	Provenance	Description
das_voyage_id	Huygens	record id; each record describes a voyage of a VOC ship
outward_return	Huygens	1 if outward voyage; 0 if return voyage
ship_name	Huygens	name of ship
ship_tonnage	Huygens	tonnage of ship
chamber	Huygens	VOC chamber that managed the voyage
departure_date	Huygens	date of departure
departure_place	Huygens	place of departure
arrival_date_cape	Huygens	date of arrival at the stopover at Cape of Good Hope
departure_date_cape	Huygens	date of departure from Cape of Good Hope
arrival_date	Huygens	date of arrival at destination
arrival_place	Huygens	destination place

Table 15. Description of variables included in file `voc_voyages.csv`

Column name	Provenance	Description
beneficiary_id	NA	record id; each record describes the beneficiary of a month letter and their relation to the VOC employee who signed the letter
full_name	NA	full name of beneficiary
first_name	NA	first name of beneficiary
patronymic	NA	patronymic of beneficiary
family_name_prefix	NA	family name prefix of beneficiary
family_name	NA	family name of beneficiary
place_of_origin	NA	place of origin of beneficiary
relation	NA	beneficiary's relation to VOC employee
remark	NA	remark on month letter and/or beneficiary relation; in case the beneficiary was an institution, this field often mentions the name of the institution
vocop_id	NA	id of muster record to which beneficiary is related, refers to file <code>VOC_persons_contracts.csv</code>
uid	NA	unique record id in database of Dutch National Archives

Table 16. Description of variables included in file `voc_beneficiaries.csv`

A.2 Data schema

The schema in Figure 5 lists all the variables included in the enriched data set (see Appendix A.1), and shows how the various files are interconnected.

A.3 Explanations of various reasons for a contract to end

Every muster record in the original data set contains an explanation of the reason why the crew member's contract ended. To avoid data redundancy, we provide the explanations of the various reasons in Table 17.

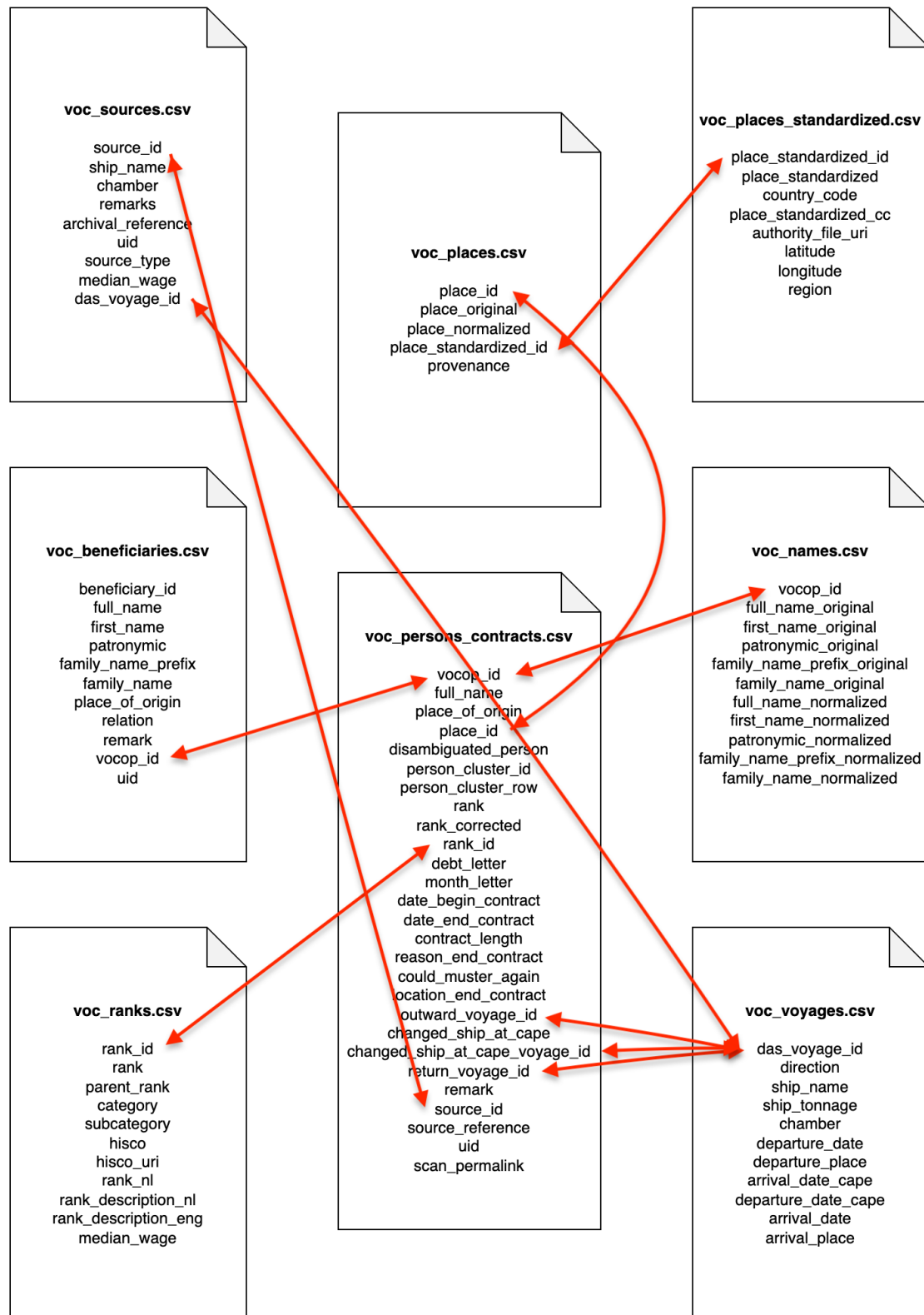


Fig. 5. Schema showing all data variables included in the seven files, and the connections between the files.

Reason for end contract	Explanation
Absent upon departure	Usually seafarers who had already received an advance (of pay in hand) but are not present at the muster prior to sailing. Some are present but are no longer suitable for the work. Others have been placed on other ships. This is noted as general remark.
Age	The seafarer receives a 'rustgage' that is a kind of pension payment or he enters an institution, for example a pauper's house.
Amsterdam chamber	Came onboard at the Cape in service with the Amsterdam chamber; thus employees who were contracted into service at the Cape for the Amsterdam chamber.
Death penalty	
Deceased	
Delft chamber	Came onboard at the Cape in service with the Delft chamber.
Deserted	When indicated where or to whom he has run away to, then this is stated under remarks.
Dismissal	The seafarer is dismissed. If further explanation is given then this is stated under 'remarks'. When a seafarer is dismissed and repatriated to Holland, then 'repatriated' is entered and 'punished' is stated as under 'remarks'.
Enkhuizen chamber	Came onboard at the Cape in service with the Enkhuizen chamber.
Free citizen	This is recorded when the seafarer's last post informs that he has become a free citizen. As a free citizen, a seafarer sometimes renews his service. When a seafarer becomes a free citizen and the date of death is known, then this is entered.
Hoorn chamber	Came onboard at the Cape in service with the Hoorn chamber.
In lening gaan	
Last record	No reason is given for ending tenure and often the account has not been closed. 'Last record' is also often used when the 'reason' remains obscure.
Missing	Not present at the muster and the reason is unknown, otherwise 'absent'.
Murdered	
Not recorded	
Otherwise	A different reason is noted under Remarks
Penalised or punished	Often involved having to work for the company while earnings were confiscated with a declaration of the pay forfeited. Death penalty is noted under Remarks. Often persons would be punished by repatriation to Holland without pay.
Remains at the Cape	
Removed	Removed from board by a foreign authority. This often happened in English ports because of suspicions that they were English; who were prohibited from signing on foreign ships.
Repatriated	Returned to the Netherlands and discharged.
Resignation	The sea-voyager resigns. Further explanation, if any, is stated under 'remarks'.
Rotterdam chamber	Came onboard at the Cape in service with the Rotterdam chamber.
Shipwrecked	This is only noted when the ship has sunk and the seafarer has drowned. It applies only to ships that were part of Dutch Asiatic Shipping. If the seafarer had been rescued while the ship was lost a different end of tenure is recorded.
To a man of war	The sea-voyager receives permission to enter service on a Dutch man of war
To a private ship	The sea-voyager receives permission to enter service on a private ship.
To regiment	
Transferred	The seafarer is transferred to another VOC ship; his pay administration is also transferred to that ship.
Unfit to work	Mostly noted as 'impotent' in ship's pay-ledgers. When those with an occupational disability are sent back to the Netherlands, they are listed as 'repatriated' and under Remarks is noted 'unfit to work'.
Unknown	
Woman	When a woman pretending to be a man is later exposed, she is fired. The ship's pay-registers give no indication as to what further eventuates.
Zeeland chamber	Came onboard at the Cape in service with the Zeeland chamber.

Table 17. Reasons for a crew member's contract to end and their explanations as provided by the Dutch National Archives.